

KODEX — ADR Cost vs. Quality (SAMPLE / SYNTHETIC DATA)

Generated 2026-06-02T17:23:04+00:00 (offline) · Procedure: Cervical Artificial Disc Replacement (DEMO DATA) · Geography: CA / 93401 within 100.0 miles.

Read this first

What this is. KODEX is a decision-support aggregator. It pulls public provider, cost, and quality-signal data, scores it with transparent and configurable weights, and prints a two-axis matrix of cost versus a labeled quality proxy, plus a summary of the published evidence on the operation.

What it is NOT. It is not a source of per-surgeon clinical success rates — those do not exist in public, machine-readable form. Every quality number here is a clearly labeled proxy or a facility-level (not surgeon-level) measure. Where a data point cannot be sourced it is shown as UNKNOWN; nothing is interpolated or invented.

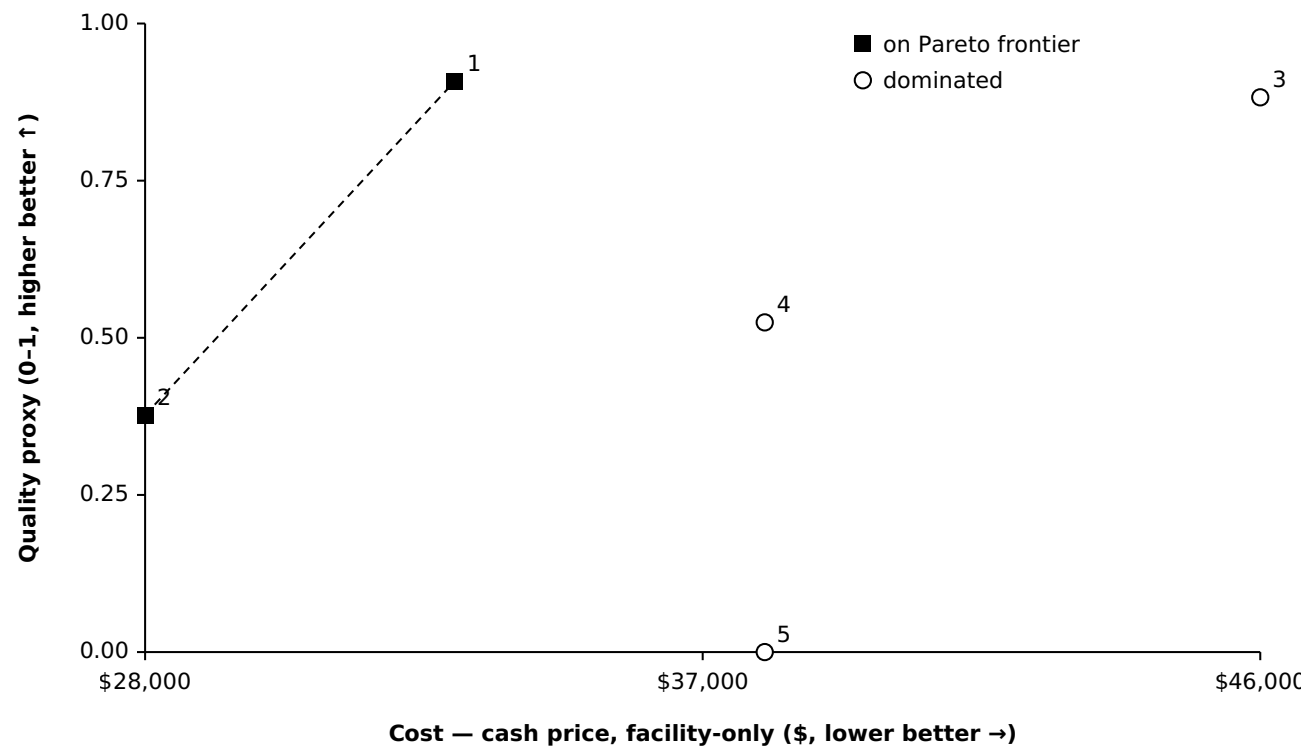
Quality axis. The "quality proxy" is a weighted blend of obtainable signals (board certification, spine fellowship, a Medicare volume floor, facility-level complication/readmission measures, and years in practice). It is a proxy, not an outcome guarantee.

Cost axis. Cost is the hospital facility cash price for the relevant CPT codes. It is not the all-in episode cost — surgeon professional fees, anesthesia, the implant device, imaging, and follow-up are extra (see the Cost Breakdown section).

This does not replace a surgical consultation or a second opinion. The strongest thing KODEX offers is a defensible shortlist of facilities and surgeons to ask about — not a verdict.

The matrix — cost vs. quality proxy

Each point is a surgeon × facility pairing. Lower-and-to-the-left is cheaper; higher-up is a stronger quality proxy. Points on the **Pareto frontier** (filled squares, connected by the dashed line) are not beaten on both axes by any other point — they are the defensible shortlist to ask about.



#	Surgeon	Facility	Cost (cash)	Quality proxy	Frontier	Data completeness
1	Ada Vela	Coast General Spine Center	\$33,000 (facility-only)	0.91	■ yes	86%
2	Cara Singh	Valley Community Hospital	\$28,000 (facility-only)	0.38	■ yes	79%
3	Dev Romero	Harbor University Medical	\$46,000 (facility-only)	0.88	○ no	86%
4	Eli Nakamura	Inland Regional Medical	\$38,000 (facility-only)	0.52	○ no	86%
5	Ben Ortega	Inland Regional Medical	\$38,000 (facility-only)	0.00	○ no	86%

Rows showing UNKNOWN for quality or cost cannot be plotted and are excluded from the frontier (they are still listed for transparency).

Per-provider detail

#1 · Ada Vela NPI 1011111111

Surgeon

- Credential (self-reported, NPPES): MD
- Taxonomy: Orthopaedic Surgery
- Board certified (ABMS, manual): True — American Board of Orthopaedic Surgery
- Spine fellowship: UNKNOWN
- License active (state board, manual): True
- Years in practice (NPPES enumeration proxy): 20
- Medicare ADR volume (FFS floor, not total): 65
- Medicare avg payment: \$9,000
- Open Payments total (transparency, display-only): UNKNOWN
- Top paying companies: UNKNOWN

Facility — Coast General Spine Center CCN 050001

- Location: CA 93401
- Cost source: MRF
- Complication measure (facility-level, not surgeon): 0.78
- Readmission measure (facility-level, not surgeon): 12.5

Quality proxy: 0.91 · completeness 86%

Proxy component	Contribution
Board certified	0.29
Spine fellowship	UNKNOWN
Medicare ADR volume (FFS floor)	0.24
Facility complication (facility-level)	0.20
Facility readmission (facility-level)	0.10
Years in practice (proxy)	0.08

#2 · Cara Singh NPI 1033333333

Surgeon

- Credential (self-reported, NPPES): MD
- Taxonomy: Orthopaedic Surgery
- Board certified (ABMS, manual): True — American Board of Orthopaedic Surgery
- Spine fellowship: UNKNOWN
- License active (state board, manual): True
- Years in practice (NPPES enumeration proxy): 10
- Medicare ADR volume (FFS floor, not total): 25
- Medicare avg payment: \$9,000
- Open Payments total (transparency, display-only): UNKNOWN
- Top paying companies: UNKNOWN

Facility — Valley Community

Hospital CCN 050003

- Location: CA 93405
- Cost source: FAIRHEALTH
- Complication measure (facility-level, not surgeon): 1.2
- Readmission measure (facility-level, not surgeon): 16.0

Quality proxy: 0.38 · completeness 79%

Proxy component	Contribution
Board certified	0.29
Spine fellowship	UNKNOWN
Medicare ADR volume (FFS floor)	0.04
Facility complication (facility-level)	0.00
Facility readmission (facility-level)	0.00
Years in practice (proxy)	0.04

#3 · Dev Romero NPI 1044444444

Surgeon

- Credential (self-reported, NPPES): MD
- Taxonomy: Orthopaedic Surgery
- Board certified (ABMS, manual): True — American Board of Orthopaedic Surgery
- Spine fellowship: UNKNOWN
- License active (state board, manual): True
- Years in practice (NPPES enumeration proxy): 25
- Medicare ADR volume (FFS floor, not total): 40
- Medicare avg payment: \$9,000
- Open Payments total (transparency, display-only): UNKNOWN
- Top paying companies: UNKNOWN

Facility — Harbor University Medical

CCN 050004

- Location: CA 93420
- Cost source: MRF
- Complication measure (facility-level, not surgeon): 0.7
- Readmission measure (facility-level, not surgeon): 12.0

Quality proxy: 0.88 · completeness 86%

Proxy component	Contribution
Board certified	0.29
Spine fellowship	UNKNOWN
Medicare ADR volume (FFS floor)	0.14
Facility complication (facility-level)	0.24
Facility readmission (facility-level)	0.12
Years in practice (proxy)	0.10

#4 · Eli Nakamura NPI 1055555555

Surgeon

- Credential (self-reported, NPPES): MD
- Taxonomy: Orthopaedic Surgery
- Board certified (ABMS, manual): True — American Board of Orthopaedic Surgery
- Spine fellowship: UNKNOWN
- License active (state board, manual): True
- Years in practice (NPPES enumeration proxy): 11
- Medicare ADR volume (FFS floor, not total): 35
- Medicare avg payment: \$9,000
- Open Payments total (transparency, display-only): UNKNOWN
- Top paying companies: UNKNOWN

Facility — Inland Regional Medical

CCN 050002

- Location: CA 93446
- Cost source: MRF
- Complication measure (facility-level, not surgeon): 1.1
- Readmission measure (facility-level, not surgeon): 15.0

Quality proxy: 0.52 · completeness 86%

Proxy component	Contribution
Board certified	0.29
Spine fellowship	UNKNOWN
Medicare ADR volume (FFS floor)	0.11
Facility complication (facility-level)	0.05
Facility readmission (facility-level)	0.03
Years in practice (proxy)	0.04

#5 · Ben Ortega NPI 1022222222

⚠ **Disciplinary action on record.** 2017 license reprimand (synthetic) (state medical board, manual check)

Surgeon

- Credential (self-reported, NPPES): MD
- Taxonomy: Orthopaedic Surgery
- Board certified (ABMS, manual): True — American Board of Orthopaedic Surgery
- Spine fellowship: UNKNOWN
- License active (state board, manual): True
- Years in practice (NPPES enumeration proxy): 14
- Medicare ADR volume (FFS floor, not total): 20
- Medicare avg payment: \$9,000
- Open Payments total (transparency, display-only): UNKNOWN
- Top paying companies: UNKNOWN

Facility — Inland Regional Medical CCN 050002

- Location: CA 93446
- Cost source: MRF
- Complication measure (facility-level, not surgeon): 1.1
- Readmission measure (facility-level, not surgeon): 15.0

Quality proxy: 0.00 · completeness 86%

Proxy component	Contribution
Board certified	0.29
Spine fellowship	UNKNOWN
Medicare ADR volume (FFS floor)	0.00
Facility complication (facility-level)	0.05
Facility readmission (facility-level)	0.03
Years in practice (proxy)	0.05
Disciplinary penalty	-0.50

Cost breakdown

The cash price below is the hospital FACILITY fee only. The full episode also includes the surgeon's professional fee, anesthesia, the implant device, pre-op imaging, and follow-up — none of which are in these numbers. Treat the facility cash price as a floor and a negotiating anchor, not the all-in cost.

#	Facility	Per-CPT cash price	Bundle total (facility-only)
1	Coast General Spine Center	22856: \$33,000	\$33,000
2	Valley Community Hospital	22856: \$28,000	\$28,000
3	Harbor University Medical	22856: \$46,000	\$46,000
4	Inland Regional Medical	22856: \$38,000	\$38,000
5	Inland Regional Medical	22856: \$38,000	\$38,000

Geographic benchmark (FAIR Health, manual entry)

CPT	Description	ZIP	Estimate	As of
22856	Total disc arthroplasty, cervical, single interspace	93405	\$28,000	2026-01

Procedure-evidence summary

This section is about **the operation**, not about any one surgeon. The findings are aggregate results from the published literature and must not be read as a prediction for a specific provider.

- **Cervical disc arthroplasty vs. fusion: a systematic review (SYNTHETIC DEMO ENTRY)** (2025) — Demo Spine Journal PMID 90000001

Across pooled cohorts, cervical ADR showed comparable or lower reoperation rates than fusion at mid-term follow-up. Figures are illustrative placeholders for this sample report.

- **Long-term outcomes of cervical total disc replacement (SYNTHETIC DEMO ENTRY)** (2024) — Demo Journal of Orthopaedics PMID 90000002

Illustrative aggregate finding only; not attributable to any individual surgeon.

Sources & dates appendix

Source	As of	Known limitation
NPES NPI Registry	2026-06-02	Credential string is self-reported free text; not board verification.
Medicare Physician & Other Practitioners	CY2024	Original Medicare FFS only. ADR patients skew younger — volume is a FLOOR, not a count.
CMS Open Payments	program-year dependent	Transparency signal only; industry ties are common and not inherently bad. Display-only by default.
Hospital Price Transparency MRFs	per-hospital	Facility component only; not the all-in episode. Schemas/compliance vary; UNKNOWN where unavailable.
CMS Care Compare	CY2024	FACILITY-level measures, NOT surgeon-level.
FAIR Health (manual)	operator-entered	Consumer benchmark; used as fallback when an MRF is unavailable.
ABMS / state board (manual)	operator-entered	One-at-a-time manual verification; blank renders as UNKNOWN.
PubMed E-utilities	2026-06-02	AGGREGATE procedure evidence only; never attributed to an individual surgeon.

Scoring weights used

Signal	Weight
board_certified	0.25
fellowship_spine	0.15
medicare_adr_volume	0.2
facility_complication	0.2
facility_readmission	0.1
years_in_practice	0.1
open_payments	0.0

To change any weight, endpoint, or the geography, the operator edits `config.yaml` and re-runs — no code changes.